Sahaj Bhadja

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Summary

Backend infrastructure engineer with 2+ years automating CI/CD, containerized deployments, and Linux build pipelines in **Python** and **Bash** across **AWS**, Azure, and **Kubernetes**, while running production LLM workloads at 99%+ reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Ran production **LLM workloads** on FastAPI background workers with explicit job-state transitions and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Operated recurring integration pipelines across PostgreSQL, SQL Server, and cloud storage with **CI/CD**-backed releases and reconciliation checks, sustaining 99%+ reliability over 5K+ monthly jobs for 500+ users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Shipped a Python and React platform through **GitHub Actions** CI/CD to Render and Vercel on AWS RDS, serving 150+ instructors and 10K+ daily requests with stable latency at peak load.
- Integrated GPT-4o, Claude, and Qwen-VL behind a switchable **LLM** provider layer with timeout fallback and per-call cost logging, plus concurrent batch workers processing 1K+ files per run.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated nightly **Jenkins** pipelines that allocated VMs, pulled commits, built executables, installed them, and ran integration and unit tests across Windows, Linux, and macOS with emailed coverage reports.
- Ran parallel builds on India and US servers with feature, staging, and production branches in GitHub, and redesigned AWS Glue and S3 **ETL**, boosting pipeline efficiency 68% and cutting storage costs 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Provisioned infrastructure with **Terraform** and deployed Docker microservices to **Kubernetes** on AWS through Jenkins CI/CD with staged approval gates and rolling zero-downtime updates, reaching 95%+ build validation.
- Secured and monitored eight Node.js services with JWT, RBAC, Snyk, and OWASP Dependency-Check, plus Sentry and Kibana **observability**, while syncing 2M+ records across locations via real-time WebSockets.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Python Flask APIs integrated with existing **AWS**-hosted authentication and geo-data microservices and Firebase real-time sync, sustaining sub-200ms latency and 40% faster confirmations across 50K+ booking records.
- Automated payment event handling with idempotent **webhooks** for refunds, disputes, and chargebacks, cutting manual work 70%, and scheduled hourly Databricks and Spark micro-batches over booking and payment events.

Skills

Languages: Python, Bash, Shell, TypeScript, JavaScript, SQL, Java
Infrastructure: Linux, Docker, Kubernetes, Terraform, ARM Templates, AWS (EC2, S3, RDS, ECS/Fargate, Glue, CodePipeline), GCP, Azure
CI/CD & Automation: Jenkins, GitHub Actions, Blue Ocean, Rolling Deployments, VM Test Automation, Nightly Builds
Observability & Security: Sentry, Azure Monitor, Kibana, Grafana, Snyk, OWASP Dependency-Check, JWT, RBAC
Backend & AI: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, LLM Orchestration, Claude, GPT-4o
Frontend & Practices: React, Next.js, Agile/Scrum, Incident Troubleshooting

Projects

Contract Opportunity Ingestion Platform | *Python, Terraform, AWS EKS, Kubernetes, Docker, GitHub Actions, Claude*

- Designed a **Terraform**-provisioned EKS platform running Python workers that ingest public federal contract opportunities, an **LLM agent** that tags them by capability, GitHub Actions deploys, and Prometheus and Grafana alerting.

Linguistix | *Python, Docker, AWS ECS/Fargate, CodePipeline, CI/CD*

- Containerized with Docker and deployed a Python REST data processing API on **AWS ECS/Fargate** with automated CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability.

FinNexus Live | *Kubernetes, Next.js, TypeScript, MongoDB, WebSockets*

- Deployed and managed a TypeScript and Next.js fintech platform on **Kubernetes** with CI/CD automation and a token-based firewall, handling real-time WebSocket updates for 50K+ users and 1M+ transactions.